CATERPILLAR®

DEALER BEST PRACTICE SERIES

CRC Certification Promotes Continuous Improvement

Application | Maintenance | **Component Rebuild** | Component Life Management | Safety | MARC Management

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May 2008
0708-4.3-1125

1.0 Introduction

Dealer Component Rebuild Centers (CRCs) are vital to the support of Mining customers in achieving machine cost per hour and availability objectives. CRCs also represent a major source of revenue income for both Caterpillar and Caterpillar dealers.

A successful CRC must achieve high standards of quality, efficiency, cleanliness, standardized processes, consistency, and image. The Global Mining CRC Certification program is a process to assess CRC performance and standards, identify opportunities for improvement, and create an integrated Continuous Improvement plan.

2.0 Best Practice Description

Today's CRC Certification Guide (copy attached to this BP) examines approximately 89 key processes with 285 standards. The standards represent the top critical success factors that significantly impact quality, efficiency, capacity, image, and administration. This ultimately affects the ability to meet customer component life, availability, and cost per ton objectives, as well as meet dealer financial goals.

1. Management - Strategic	2. Operational Processes	3. Facility / Tools	4. Receiving / Pre-clean	5. Disassembly / Cleaning
Metrics & KPIs - Mgt Reports	Work Flow Process Mapping	Material handling and staging	Area sizing / access	Area Sizing
CI process	Parts / Supplies Delivery	Safety equipment	Staging - Area	Staging - Baskets/racks/stand
Service Administration	Parts / Reuse	Tool management	Receiving Inspection	Control Doc's & Checklists
Technical Communications	Reman parts use process	Work Flow	Material Handling	Disassembly/ Inspect process
Service training program	Communications		Clean component policy	Cleaning process
Shop safety process	Tool maint./calibration process		Material Handling	Shop Equipment
Failure analysis process	Contamination Control Process			Power Cleaning
Standard jobs mgt process	Housekeeping Policy			Parts ordering
Flat rate mgt. process				
6. Parts Salvage	**7. Component Rebuild**	**8. Assembly - General**	**9. Test & Adjust**	**10. Painting/Shipping/R&I**
Facility (dealer-owned)	Area Sizing	Area Sizing	Area Sizing	Area Sizing
Salvage processes	Staging - Baskets/racks/stand	Staging - Baskets/racks/stand	Staging - Area	Staging - Area
Parts managing	Control Doc's & Checklists	Control Doc's & Checklists	Control Doc's & Checklists	Painting process
	Parts, Materials & Supplies	Parts, Materials & Supplies	Test cell / area	R & I process
	Rebuild Process	Assemble process	Dynamometer test	Shipping
	Shop Equipment	Assemble Tools	Bench test	
		Shop Equipment		

The CRC Certification Guide is designed to assess each of the 89 key processes by checking the enabling standards. The standards include:

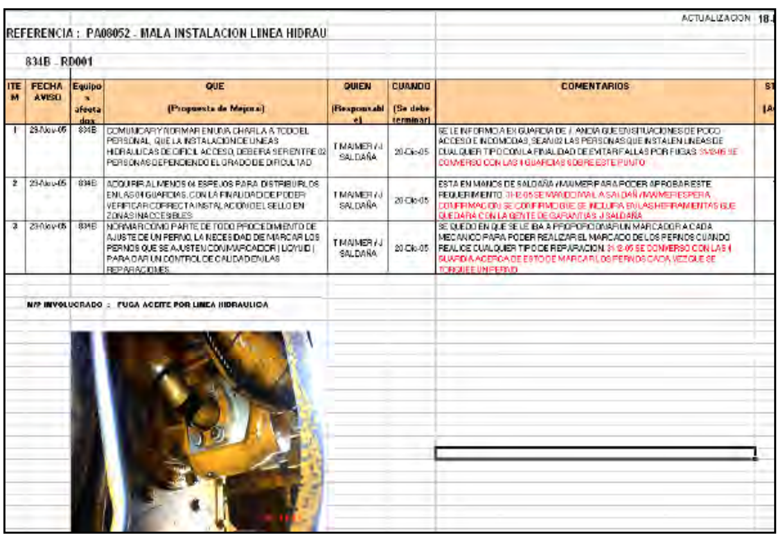
- Does the process exist?
- Is there a process owner?
- Is the process well implemented and kept current?
- Is the process followed?
- Does the process produce the expected result?

Example Process and Standards

01.07	Failure analysis process	a AFA process owner		
		b Qualified technician (AFA I, II, III)		Are courses conducted by standard dealer, regional dealer or Cat center
		c Designated AFA area		Suggest AFA area be near disassemble and cleaning area
		d Lab tools		Include: ref's, digital close-up camera, lighting, magnification, micrometers
		e Standard report form		Template provide organized records and prompt effective FA steps
		f Database for FA storage/retrieval		Electronic record are more effectively organized easier to organize

Often, there is a published Caterpillar Global Mining "Best Practice" to support a key process, as illustrated below:

Best Practice: <u>Failure Analysis Process - A Key Tool To Manage Quality</u>



ACTUALIZACION 18

REFERENCIA : PA08052 - MALA INSTALACION LINEA HIDRAU

834B - RD001

ITEM	FECHA AVISO	Equipos afectados	QUE (Propuesta de Mejora)	QUIEN (Responsable)	CUANDO (Se debe terminar)	COMENTARIOS
1	29-Nov-05	834B	COMUNICAR Y NORMAR EN UNA CHARLA A TODO EL PERSONAL, QUE LA INSTALACION DE LINEAS HIDRAULICAS DE DIFICIL ACCESO, DEBERA SER ENTRE 02 PERSONAS DEPENDIENDO EL GRADO DE DIFICULTAD	T MAIMER / J SALDAÑA	20-Dic-05	SE LE INFORMO A EX GUARDIA DE J. ANDIA QUE EN SITUACIONES DE POCO ACCESO E INCOMODAS, SEAN 02 LAS PERSONAS QUE INSTALEN LINEAS DE CUALQUIER TIPO CON LA FINALIDAD DE EVITAR FALLAS POR FUGAS. 31-12-05 SE CONVERSO CON LAS 4 GUARDIAS SOBRE ESTE PUNTO
2	29-Nov-05	834B	ADQUIRIR AL MENOS 04 ESPEJOS PARA DISTRIBUIRLOS EN LAS 04 GUARDIAS, CON LA FINALIDAD DE PODER VERIFICAR CORRECTA INSTALACION DEL SELLO EN ZONAS INACCESIBLES	T MAIMER / J SALDAÑA	20-Dic-05	ESTA EN MANOS DE SALDAÑA / MAIMER PARA PODER APROBAR ESTE REQUERIMIENTO. 31-12-05 SE MANDO MAIL A SALDAÑ / MAIMER ESPERA CONFIRMACION. SE CONFIRMO QUE SE INCLUIRA EN LAS HERRAMIENTAS QUE QUEDARA CON LA GENTE DE GARANTIAS J SALDAÑA
3	29-Nov-05	834B	NORMAR COMO PARTE DE TODO PROCEDIMIENTO DE AJUSTE DE UN PERNO, LA NECESIDAD DE MARCAR LOS PERNOS QUE SE AJUSTEN CON MARCADOR (LIQUID) PARA DAR UN CONTROL DE CALIDAD EN LAS REPARACIONES	T MAIMER / J SALDAÑA	20-Dic-05	SE QUEDO EN QUE SE LE IBA A PROPORCIONAR UN MARCADOR A CADA MECANICO PARA PODER REALIZAR EL MARCADO DE LOS PERNOS CUANDO REALICE CUALQUIER TIPO DE REPARACION. 31-12-05 SE CONVERSO CON LAS 4 GUARDIA ACERCA DE ESTO DE MARCAR LOS PERNOS CADA VEZ QUE SE TORQUEE UN PERNO

N/P INVOLUCRADO : FUGA ACEITE POR LINEA HIDRAULICA

The report form, database, lab, qualified technician, tooling, and communication equipment all contribute to an effective Applied Failure Analysis process/system.

The objectives of CRC Certification are to assist dealers in identifying improvement opportunities and developing work plans for CRC Continuous Improvement.

Key areas include:

Quality
Component rebuild quality not only affects warranty costs, it also provides expected component life and machine availability to mining customers. These contributions have significant impact on MARC contract financial performance and ultimately impacts new machine sales. Quality rebuilt components influence customer machine, parts, and service sales and build loyalty, relationships, and preference for Caterpillar machines and dealer product support. Certification reviews quality systems such as calibration, equipment, control documents, reuse practices, and training.

Efficiency
Parts and labor efficiency is vital to a CRC operation in reducing the costs of goods sold, direct operating costs, and component rebuild turnaround time through high utilization of CRC assets. Certification reviews bottlenecks, delays, non-value-adding processes, and other sources of waste. Also considered is equipment, job controls systems, parts reuse.

Capacity
Caterpillar active mining machine population around the world has doubled in the past three years. The population is forecast to double again in the next three years. The dealer has three basic options in providing for the growing customer's component renewal requirements:

- Become more efficient
- Invest in more facilities, tooling, and people.
- Use Reman components for peak shaving.
- Combination of the above.

Certification assessment reviews efficiency, product velocity, and asset utilization to help guide investment decisions for the most effective benefit. Infrastructure sizing and personnel development is also considered for effective throughput.

Image
Many of the Contamination Control standards are a part of CRC Certification as they impact quality and efficiency. Image is more than just a "clean" shop. It includes orderly parts staging, proper tools/equipment, smooth workflow, etc. Image certainly helps sell dealer services to your customers. Cleanliness and organization creates a professional work environment for the CRC technicians.

Administration
Administration impacts everything that happens in the shop from planning and scheduling, to standard rebuild times, flat rates, rebuild kits, technical communication, software systems, etc. Effective administration is vital to CRC performance. Certification reviews those administrative processes/systems, beyond measurement, that effect on-floor efficiency and quality.

Continuous Improvement

In 1999, Caterpillar Global Mining introduced the CRC Certification program. The intention was to provide direct Caterpillar assistance to dealers to evaluate and improve CRC operations. Since that time, over 20 Mining dealers have conducted CRC Certification reviews.

Other changes over the recent years included the creation of CGM “Best Practices” to share ideas and proven solutions, and to support the Certification process.

The CRC Certification Guide has also changed over the years to be much more process focused.

A Dealer CRC Managers “Best Practices” Knowledge Network was created as a tool for dealers to communicate with each other. Worldwide CRC Conferences are sponsored on a regular basis and conducted at a selected host dealer in order to share ideas, Best Practices, and common concerns.

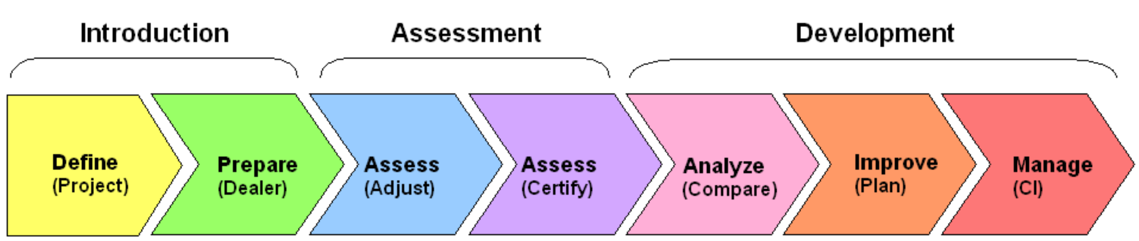
Many dealers have established a 6Sigma Blackbelt or CI Champion position with responsibility for CRC development.

These programs are all aimed at assisting dealer CRCs to provide globally consistent support to mining customers.

3.0 Implementation Steps

A CRC Certification review for all major component areas is typically performed in one week.

- The assessment is performed on the shop floor typically with a Caterpillar Global Mining team and CRC area supervisors. This usually takes three to four days.
- After the assessment, the Caterpillar/Dealer team reviews the results for consensus.
- Agreed-upon results are then analyzed to create improvement/priority recommendations.
- The dealer then determines an action plan to implement selected recommendations.



Steps to arrange / conduct a CRC Certification assessment

- Review CRC Certification Guide and Instructions
- Determine desired timeframe to perform the process phases
- Arrange for Caterpillar involvement through Regional Mining Service Rep and District SOR
- Conduct a self-assessment using the CRC Certification Guide
- Prepare CRC as needed for the certify assessment (usually 2-3 months). Document pre-assessment adjustments
- Conduct assessment, analyze findings, develop recommendations, and create action plan.
- Implement action plan.

4.0 Benefits

Most dealers completing the CRC Certification have developed a variety of improvement recommendations ranging from simple “Quick Wins” to significant “Process” changes. Several of the CRC Manager comments, regarding the “value” of the Certification processes, are as follows:

“ We started tracking our average engine turnaround time and found it was over 35 days. Changes we have made have taken out over 5 days”

“We eliminated almost two technicians from the piston, rod, and liner rebuild area and redeployed them. We will save $250,000 per year.”

“We changed our administrative process and reduced average component turnaround time by five days.”

“We can get one additional life out of engine blocks by installing inserts at the first rebuild”

5.0 Resources Required

- Management time to prepare and conduct assessment
- Staff time to implement work/action plan
- Incremental improvement investment costs

6.0 Supporting Attachments / References

CRCC Guide 08.04.20.xls

7.0 Related Best Practices

Failure Analysis Process – A Key Tool to Manage Quality

8.0 Acknowledgements

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